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Genetic Polymorphism

Hardy-Weinberg Equilibrium

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What is Genetic Polymorphism

- Genetic Polymorphism is the variation in genetic material
- **Polymorphic traits are discreet traits that are controlled by single genes and are less influenced by environment**
- Ex. The ability to taste PTC / ABO Blood Group
- **Genetic Polymorphism is the occurrence of more than one allele for a particular trait**
- A trait is called a polymorphic trait only when the frequency of the gene is **more than 1% in the population**

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What is Genetic Polymorphism

Sources of Genetic Polymorphism

- Genetic polymorphism is in fact the manifestation of the process of evolution
- The factors responsible for GP are the evolutionary processes like Mutation and Migration

Levels of Genetic Polymorphism

- Genetic Polymorphism exists at multiple levels
- These levels include cells (ABO), chromosomes (X and Y) , proteins (Hba & Hbs) etc

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Polymorphism & Selection

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Genetic Polymorphism & Natural Selection

- The role of natural selection and its relation to GP is unclear
- Debate exists on whether all the polymorphism exhibited by man has any selective value in evolution of this species – Like ABO?
- Some polymorphic traits are very important for the survival of the individual itself - like for example, Sex, which is a **permanently polymorphic trait**
- Scientists like Kimura say that a majority of polymorphic traits have no role to play vis-à-vis natural selection and such phenomenon is termed **"Selectively neutral polymorphism"**

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Genetic Polymorphism & Natural Selection

- However, genetic polymorphism is important for **maintaining the heterozygosity in the population.**
- If all the genes are monomorphic, then there **is little biochemical diversity** existing at the gene loci
- Such a situation will help individuals survive as long as the environment is stable
- However, if the environment changes, stabilizing natural selection is replaced either by directional selection or disruptive selection and natural selection has little choice to select the individuals with any "adaptive advantages"
- Heterozygosity is a precondition for variation, hence evolution
- Hence, genetic polymorphism is a **predisposition of the population to adaptation**

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Balanced Genetic Polymorphism

- **It is a process by which natural selection actively maintains multiple alleles in a population at a frequency above that of gene mutation**
- This usually happens when the heterozygotes have higher adaptive significance than the homozygotes – a condition called **heterozygote advantage or heterotic balanced selection**
- Ex. Sickle Cell Trait (HbA Hbs) – A balance between selection against homozygous sufferers and selection against homozygous normal

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Genetic Polymorphism & Natural Selection

- **In directional selection**, the allele frequency for a trait continuously shifts in one direction, thus favoring a particular phenotype
- **In stabilizing selection** the frequency of the alleles of lower fitness decreases until they vanish
- **Balancing selection** is similar to disruptive selection where individuals of extreme trait values are favored against those with average trait values.

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Mendelian Population

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Mendelian Population

- Population means a spatial and temporal group of conspecific, interbreeding individuals
- Population Geneticists, however, further restrict the definition to a group of genes rather than a group of individuals and call it a **GENE POOL**
- A Gene Pool is the total collection of all the allelic forms that are carried by all the gametes of a population
- A gene pool has been variously termed as Panmixia by Sewall Wright, Mendelian Population by Dobzhansky and Gamedeme by Glimour and Gregor
- The term that is most common today is DEME

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Mendelian Population

- A deme represents a local population of interbreeding or potentially interbreeding individuals i.e., any gamete in this population has an equal chance to unite with any gamete of the opposite sex in the same population and result in a viable zygote
- Successful reproduction in a population however is prevented by a number of factors
- Just like in the case of an individual, we can also calculate the phenotype and genotype of a population through statistical methods
- Understanding the behavior of genes and application of Mendelian principles to the population is the specialization of **POPULATION GENETICS**

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Population Genetics

1. To understand how populations maintain **heterozygosity**
2. **Human genetics** – since man is not amenable to genetic experiments, population genetics provides an indirect way to understand the distribution and dynamics of human genes
3. Man is not a unit of **evolution**. Hence, population genetics will provide a realistic model to understand evolution at a population level – i.e., how gene frequencies are maintained or changed in a population
4. Spin off – development of sophisticated **mathematical methodologies** to explain how allelic frequencies can be calculated

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Hardy Weinberg Equilibrium

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Hardy-Weinberg Equilibrium

- Population Genetics is primarily interested to know how genes behave in a population since the latter is the unit of evolution
- To study the dynamics of the alleles in changing population, a reference model was provided by Godfrey Hardy, a mathematician and Wilhelm Weinberg, an experimental biologist (1908)
- The Hardy-Weinberg model represents a STATIC GENETIC POPULATION, the one in which genes undergo no change and genetic frequencies are in a state of equilibrium (Hardy-Weinberg Equilibrium)

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Hardy Weinberg Equilibrium

- ***In a panmictic (randomly mating) population, without disruptive influences, the frequencies of alleles of a gene remain unchanged in every generation.***
- ***But if this equilibrium is disturbed in some manner, the genotype frequencies reach a new equilibrium value in one or more generations, depending on the nature of the disrupting influence.***
- Though such an ideal population where the allelic frequencies are static hardly exists, it provides the fundamental model against which the actual gene frequencies can be measured

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Hardy-Weinberg Equilibrium

- A population is said to be evolving if the frequencies of its alleles are changing; it is not evolving if these frequencies remain constant
- The latter situation is genetic equilibrium
- Certain conditions must be met for the population to be in Genetic Equilibrium
  1. **Random Mating**
  2. **Large Population Size**
  3. **Lack of Evolutionary Forces**
  4. **All Matings Should Be Equally Fertile**

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Hardy Weinberg Equilibrium

- The algebraic representation of H/W Equilibrium is

$$p^2 + 2pq + q^2 = 1$$

  - **p = [A] i.e., the frequency of dominant allele**
  - **q = [a] i.e., the frequency of recessive allele**
  - **'A' and 'a' are the allelic forms of one gene**

- This equation represents a hypothetical situation of no change when certain condition are met

- This static model of H/W is in fact the allelic occurrence seen in the classic Mendelian inheritance of phenotypic ratio of 3:1

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Factors disturbing HWE

- **Mutations** – adding new / novel alleles to population
- **Migration and Isolation** – adding / preventing new alleles to a population
- **Genetic Drift** – causing massive shifts in allelic frequencies
- **Selection** – favoring certain special allelic combinations over others in specific environment
- **Inbreeding** – reducing heterozygozity and increasing homozygozity in the population

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Non Random Mating

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Non Random Mating

- **Genetic recombination does not by itself alter allele frequencies**
- However, any consistent bias in mating patterns can alter the genotypic proportions
- By affecting the **combination frequencies of genotypes**, nonrandom mating causes deviations from the Hardy Weinberg expectations
- It therefore sets the stage for the action of natural selection
- Basically, there are three models of nonrandom mating – positive assortative mating, negative assortative mating and inbreeding

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Positive Assortative Mating

- This occurs when individuals of like phenotype mate more often than expected by random mating predictions
- Because individuals with like phenotypes are also similar to some degree in genotypes as well, the result of positive assortative mating increases the amount of homozygosity in the population and reduces heterozygosity
- The values of P^2 and Q^2 are greater than expected; and value of 2pq is less than expected
- Ex. Stature and IQ or Eye color

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Negative Assortative Mating

- This involves mating with an individual who is phenotypically dissimilar
- Theoretically, if this occurs more than expected by random mating predictions, it should increase the amount of heterozygosity in the population while correspondingly reducing homozygosity

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Inbreeding

- Inbreeding, also called **genetic consanguinity**, occurs when relatives mate more often than expected
- Such mating will increase the amount of homozygosity, since relatives who share close ancestors will more than likely also share similar genes
- Inbreeding occurs less frequently than predicted under random mating, because all societies have some form of incest taboo, esp. nuclear family
- Inbreeding reduces the variability among the offspring thereby reducing the reproductive fitness

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Inbreeding

- In some societies, due to the small number of potential mates available, inbreeding among fairly close relations such as cousins is actively encouraged or is unavoidable Ex. Pitcairn Island & Tahitian wives (four families, 50 inhabitants as of 2010)
- There are some areas, on the other hand, where inbreeding is avoidable but it is still actively encouraged
- Japan – First Cousin Marriage – 10% of all marriages
- AP, India – Uncle – Niece Marriage – 10% of all marriages

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Inbreeding

- However, such considerable inbreeding is the exception rather than the rule among human populations and most groups actively work very hard at maintaining exogamy
- As a general rule, then, most human populations do not inbreed if they can at all help it
- Inbreeding has important medical consequences in addition to its effect on genetic equilibrium
- When relatives mate, their offspring have increased probability of inheriting an allele in homozygous dose

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Genetic Effects of Consanguineous and Non-Consanguineous mating

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Genetic Effects of Consanguineous and Non-Consanguineous mating

- Any departure from random mating naturally leads to complications in the relationships between allele frequencies and genotype frequencies
- If a heterozygote Aa is self-fertilized, it will produce three kinds of progeny AA, Aa and aa, in the ratio of 1:2:1
- At this stage, the frequency of the heterozygote is 0.5
- If self fertilization is continued for another generation, the homozygotes will breed true but the heterozygotes will again segregate, reducing their frequency to 0.25

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Genetic Effects of Consanguineous and Non-Consanguineous mating

- With every succeeding generation of self fertilization, the frequency of the heterozygotes drops off by 50%, reaching 0.008 by generation seven and 0.001 by generation ten.
- At this stage, the population is 99.9% homozygous
- The process of repeated self fertilization illustrates the general effects of a form of nonrandom mating called Inbreeding
- Inbreeding occurs when the mates are genetically related

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Genetic Effects of Consanguineous and Non-Consanguineous mating

- Self fertilization is the extreme example, but other examples such as the mating of full siblings, first cousins, parents and offspring, and half siblings have the same effect, namely, to increase the frequency of homozygotes and decrease the frequency of heterozygotes.
- As a result, the H/W method does not hold
- Another genetic effect of inbreeding is that ***two alleles in an individual will be identical by descent from a common ancestor***

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Genetic Load

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What is Genetic Load

- Genetic Polymorphism is not necessarily an unqualified boon to the population
- The vast genetic diversity provides raw material for evolution and also equips genetic predisposition to adaptability in populations
- But obviously it also includes several alleles which do not impart optimum fitness to the population and at times may be harmful to it
- Such deleterious genes are a genetic liability which tends to decrease the fitness of the population and is known as Genetic Load
- According to Crow *"Genetic Load is the proportion by which the fitness of an average genotype in the population is reduced in comparison with the best genotype"*

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Types of Genetic Load

Three major types of genetic loads originate and maintain in a population

- Mutational Load
- Segregational Load
- Incompatibility Load

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Mutational Load

- Mutational load involves the slicing away of genetic variability due to recurrent mutations
- Along with deleterious genes, even the beneficial genes may also be eliminated through mutations leading to genetic burden on the population
- If “a” is beneficial but is converted to “A” because of recurrent mutations, the frequency of “a” will reduce and that of “A” is increase, thereby increasing the genetic load

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Segregational Load

- It is known that heterozygotes are usually more successful in efficient living than the homozygotes
- Segregational load refers to a process where inferior homozygotes are continuously segregated
- These homozygotes represent a liability in the population because the population carries these homozygotes without any immediate and obvious utility
- Segregational load can cause greater lowering of variability then the mutational load

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Incompatibility Load

- Incompatibility load is encountered when certain genotypes are unable to survive in the environment of specific parental genotypes
- Ex. Erythroblastosis Foetalis

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